VENKATA KARTHIK VADLAMUDI

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SUMMARY

Software Engineer with 3+ years building distributed systems and LLM applications at production scale. Java/Python backend specialist with hands on experience in high availability system design, observability, and full SDLC ownership across millions of users; MS in CS from NYU (AI/ML focus); applied experience building agentic AI workflows, RAG & multimodal AI systems

EDUCATION

New York University, Tandon School of Engineering — New York, USA
Master of Science, Computer Science — May 2025

Indian Institute of Technology — Dharwad, India
Bachelor of Technology, Computer Science and Engineering — May 2021

WORK EXPERIENCE

Software Developer — June 2026 – Present
Uplink Works LLC — *Remote, United States of America*

- Shipped **LLM powered features** on **FastAPI & Spring Boot** microservices, cutting **p95 latency by 35%**
- Built **agentic AI workflows** using **tool calling & MCP** to automate multi step tasks, reducing manual work by **60%**
- Engineered a multi model **LLM gateway** with routing, semantic caching & fallbacks, lowering inference costs by **40%**
- Led code reviews and set up **LLM evals** & guardrails as **CI/CD** gates on **Kubernetes**, catching regressions pre release

AI Engineer — Sept 2025 – June 2026
Four Paws of Hope Inc. — *Remote, United States of America*

- Architected Fullstack Mobile/Web applications using **React, Node.js, Firebase & multimodal AI (Python) systems**
- Built VLMs with Semantic Search, achieving **85%+ mAP** for 1M+ images with <200ms latency in lost pets matching
- Implemented **agentic RAG workflows** with multimodal reasoning & confidence scoring, reducing false positives by **45%**
- Deployed production inference pipeline processing concurrent queries through **visual feature extraction** and **retrieval**

Software Engineer — June 2021 – May 2023
AirAsia Tech (AirAsia Technology Centre India Pvt. Ltd.) — *Bengaluru, India*

- Developed and deployed scalable **REST APIs** and **microservices** powering high volume booking and payment workflows
- Optimized app load time by **one third,** using backend **Java** optimizations, **SQL** query tuning, Redis caching, and selective React lazy loading, also reducing bundle size by **∼25%**, cutting drop offs by **∼20%** and raising user satisfaction by **40%+**
- Led a team of 4 frontend developers to revamp UI/UX on Web and App for two LOBs, lifting CTR by **40%** and **30%**
- Helmed 5+ end to end feature rollouts, driving a **15%+ surge** in user engagement and **10% boost** in feature adoption
- Debugged and resolved **50+** critical P0/P1 issues post release, ensuring **99.9% uptime** and rapid recovery during incidents

Software Development Intern — Sept 2020 – Jan 2021
BlynkSpeak Solutions — *Dharwad, India*

- Developed VoIP applications using Hybrid frameworks such as React Native & Flutter; Integrated with **AWS** and **Firebase**

RELEVANT PROJECTS

FinRegAI: FinTech AI Compliance Pipeline: *Python, FastAPI, LlamaIndex, Ollama, Qdrant*

- Built an asynchronous FastAPI pipeline using LlamaIndex to process a self curated dataset of **300+ regulatory PDFs**, executing **token boundary chunking** to isolate precise compliance clauses for vector embedding generation
- Developed jurisdiction & risk tier **payload filters** in Qdrant, avoiding full index scans to keep retrieval under **15ms**

Cloud Academic Paper Summarizer: (GitHub) *AWS, LLMs, Hugging Face Transformers, Flask, React*

- Co led a team of 8 to build a summarizer using a RAG pipeline with various LLMs and Pinecone vector search, reducing query latency by **∼30%** and achieving a best **BERTScore of 0.89** via **prompt engineering** & model benchmarking
- Built and deployed a scalable Flask backend with AWS Lambda, DynamoDB, and API Gateway, achieving **99.8% uptime** and response times around 1.2s during peak loads of 10k concurrent users (5× baseline), with a user friendly React UI

RELEVANT SKILLS

Languages/Databases	Python, Java, TypeScript, JavaScript, C++, PostgreSQL, MySQL, NoSQL, DynamoDB, Redis
Artificial Intelligence	Generative AI, LLMs, RAG, AI Agents, ML, Computer Vision, PyTorch, LangChain, LangGraph
App/Web Frameworks	Spring Boot, FastAPI, React, Node.js, REST APIs, gRPC, React Native, Flask, Django, Flutter
Distributed/Cloud	AWS, GCP, Docker/Kubernetes, Microservices, System Design, CI/CD, Firebase, Spark, Hadoop
Tools and Technologies	Git, Linux, Hugging Face, Ollama, Grafana, Kibana, JUnit, Jest, Postman, CircleCI, Jira, Figma